THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS CHAPTER TEST SET B

Chapter 9: **Symmetry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The line along which a figure is folded to match is its: 1

(a) diagonal **(b)** line of symmetry **(c)** tangent **(d)** radius

2. How many lines of symmetry does a circle have? 1

(a) 360 **(b)** 4 **(c)** 1 **(d)** infinitely many

3. Which letter has a vertical line of symmetry? 1

(a) F **(b)** G **(c)** A **(d)** R

4. In a reflection, the distance of a point from the mirror line equals the distance of its image from: 1

(a) the origin **(b)** the mirror line **(c)** nothing **(d)** the next point

5. The order of rotational symmetry of an equilateral triangle is: 1

(a) 2 **(b)** 3 **(c)** 1 **(d)** 6

6. The order of rotational symmetry of a rectangle (not a square) is: 1

(a) 1 **(b)** 3 **(c)** 4 **(d)** 2

7. The letter N has: 1

(a) a horizontal line of symmetry **(b)** a vertical line of symmetry

(c) rotational symmetry of order 2 **(d)** order 4

8. A parallelogram (not a rectangle) has: 1

(a) 4 lines **(b)** 1 line

(c) 2 lines **(d)** no line of symmetry

9. The word MOM written vertically on a vertical mirror reads: 1

(a) MMO **(b)** MOM **(c)** OMO **(d)** WOW

10. The line of symmetry of an isosceles triangle passes through: 1

(a) no point

(b) the vertex between equal sides and the midpoint of the base

(c) only the base

(d) two base points

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Find the angle of rotational symmetry of a regular hexagon. 2

12. What is the order of rotational symmetry of an equilateral triangle? 2

13. Draw the lines of symmetry of a square. 2

14. A figure has 5 lines of symmetry. Name a regular figure with this property. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Find the order of rotational symmetry for the letters N, S, Z, X. 3

16. Draw an isosceles triangle and mark its line of symmetry. Explain why there is only one. 3

17. Does a rhombus have line symmetry? How many lines and where? 3

OR

Give three examples of symmetry in nature or architecture. 3

18. A figure has rotational symmetry of order 3. What is the smallest angle of rotation? 3

OR

How many lines of symmetry does a regular octagon have? What is its angle of rotation? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Explain reflection with the mirror-line property. 5
(ii) Draw the reflection of triangle ABC about a vertical line.
(iii) Verify equal distances.

OR

(i) How can you create a symmetrical pattern using paper folding? 5
(ii) Make a four-fold design.
(iii) Count its lines of symmetry.

20. **Case study:** The Ashoka Chakra has 24 spokes. 5
(i) Order of rotational symmetry?
(ii) Angle of rotation?
(iii) Lines of symmetry?
(iv) Why is the wheel considered symmetrical?

OR

(i) Tabulate the lines and order of symmetry for rectangle, rhombus, parallelogram and kite. 5
(ii) Give reasons.

END OF QUESTION PAPER — Total: 40 Marks

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